

# ReadMe: A Persuasive Mobile App for Encouraging Reading Among Children

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## Setup documentation

Choose ONE:

- Option 1 (fastest): install the APK and use the app.
- Option 2: do a full setup in your own Firebase project (you will not be added to the original Firebase project).

### **Option 1: Install the Android APK and use the app (fastest)**

APK path (relative to the project root, after building):

- build/app/outputs/flutter-apk/app-release.apk

Steps:

1. Build the APK (skip this if the APK already exists): flutter build apk --release
2. Use the generated APK at: build/app/outputs/flutter-apk/app-release.apk
3. Copy it to an Android device and install it (you may need to allow "install unknown apps").
4. Open the app and sign in or create an account.

### **Option 2: Full setup (new Firebase project)**

Continue with the sections below to create your own Firebase project, connect the app, deploy Cloud Functions, and run the app.

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## 1. Requirements

Software prerequisites:

- Flutter SDK 3.x

- Verify: flutter --version
  - Check setup: flutter doctor
- Node.js 20+ (Cloud Functions requirement for this repository)
  - Verify: node --version
- Firebase CLI
  - Install: npm install -g firebase-tools
  - Verify: firebase --version
- FlutterFire CLI
  - Install: dart pub global activate flutterfire\_cli
  - Verify: flutterfire --version

Optional (recommended for web and CORS):

- Google Cloud SDK (gcloud and gsutil)

Accounts:

- Firebase account (required only for Option 2)
- OpenAI account and API key (required only for Option 2)

## 2. Full setup (new Firebase project)

A) Create a Firebase project (Firebase Console) Enable:

- Authentication (Email/Password)
- Firestore Database
- Storage
- Cloud Functions
- Hosting (optional)

B) Connect Flutter app to Firebase

1. firebase login
2. flutterfire configure
3. Select your new Firebase project and platforms

C) Verify configuration

- Confirm projectId and storageBucket in lib/firebase\_options.dart

## 3. Backend (Cloud Functions) setup

1. Install dependencies

- cd functions
- npm install
- cd ..

## 2. Configure OpenAI API key (Functions Secrets)

- `firebase functions:secrets:set OPENAI_KEY`
- `firebase functions:secrets:list`

## 3. Deploy

- `firebase deploy --only functions`

## 4. Verify

- Check Firebase Console: Build -> Functions
- Use the health check endpoint shown in the Functions console

# 4. Storage CORS setup (required for Web)

This step is required for the web app to load PDFs/images from Firebase Storage without browser CORS errors.

1. Confirm `cors.json` exists in project root.
2. Install Google Cloud SDK.
3. `gcloud auth login`
4. `gcloud config set project YOUR_PROJECT_ID`
5. Apply CORS:
  - `gsutil cors set cors.json gs://YOUR_STORAGE_BUCKET_NAME`
6. Verify:
  - `gsutil cors get gs://YOUR_STORAGE_BUCKET_NAME`

# 5. Running the application (development)

Web:

- `flutter run -d chrome`

Android/iOS:

- `flutter devices`
- `flutter run -d`

Windows:

- `flutter run -d windows`

# 6. Building and deployment

Build:

- Android APK: `flutter build apk --release`
- Web: `flutter build web --release`
- Windows: `flutter build windows --release`

Deploy Web Hosting (optional):

1. flutter build web --release
2. firebase deploy --only hosting

## 7. Troubleshooting

Firebase initialization errors:

- flutter clean
- flutter pub get
- flutterfire configure

Cloud Functions issues:

- firebase use
- firebase functions:secrets:list
- firebase deploy --only functions

Web CORS errors:

- gsutil cors set cors.json gs://YOUR\_STORAGE\_BUCKET\_NAME

PDF does not load:

- Confirm pdfUrl exists in the Firestore book document
- Confirm Storage object exists and is readable
- For web, confirm CORS has been applied